AP CALCULUS AB

Unit 1 • Self Check 1.4 • Lessons 1-7 to 1-9 • Answer key

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Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

f is differentiable, with $f(0) = 1$ and $f(3) = 10$. The MVT guarantees a c in $(0, 3)$ with $f'(c) =$

(A) 9

(B) 3

(C) $\frac{10}{3}$

(D) $\frac{1}{3}$

Solution

Answer: (B). $\frac{10-1}{3-0} = 3$.

Question 2

NO CALCULATOR

Where is $f(x) = \sqrt[3]{x}$ continuous but not differentiable?

(A) $x = 1$

(B) $x = 0$

(C) $x = -1$

(D) nowhere

Solution

Answer: (B). Vertical tangent.

Question 3

NO CALCULATOR

f is continuous, $f(1) = 2$ and $f(4) = 8$. Which theorem guarantees $f(c) = 5$ for some c in $(1, 4)$?

(A) MVT

(B) EVT

(C) IVT

(D) None of them

Solution

Answer: (C). A height between $f(1)$ and $f(4)$: the IVT.

Question 4 (Free response) NO CALCULATOR

A differentiable function has $f(1) = 2$, $f(3) = 8$, $f(6) = -1$. Which theorem guarantees each, and why? (a) $f(c) = 5$ for some c in $(1,3)$. (b) $f'(c) = 3$ for some c in $(1,3)$. (c) $f'(c) = -3$ for some c in $(3,6)$.

Solution
(a) IVT ($2 < 5 < 8$, continuous). (b) MVT ($\frac{8-2}{2} = 3$). (c) MVT ($\frac{-1-8}{3} = -3$).